Optimizing Machine Learning Models for Embedded Systems: A Comparative Study on Conventional PCs and Raspberry Pi

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Abstract Chronic kidney disease (CKD) is a major global health problem, and early diagnosis is important for timely treatment. This work studies whether machine learning models for CKD prediction can be deployed efficiently on a resource-constrained embedded platform. Using the publicly available kidney_disease.csv dataset, the data were cleaned, encoded, normalized, and split into training and testing sets. Logistic regression, decision tree, and neural network models were considered, and performance was examined in terms of classification accuracy, runtime, and memory usage on a conventional PC and a Raspberry Pi. The reported results show that logistic regression achieved 87% accuracy with a ROC-AUC of 0.91, while the neural network experiments reported 87% accuracy on PC and 89% on Raspberry Pi with different runtime and memory costs. The comparison shows that embedded deployment is feasible, but model selection must balance predictive performance against computational cost.

Keywords- chronic kidney disease, machine learning, embedded systems, Raspberry Pi, logistic regression, neural network

I. Introduction

Chronic kidney disease affects millions of people worldwide, and delayed diagnosis can reduce the chance of effective treatment. Conventional diagnostic workflows often depend on clinical infrastructure that may not be available in remote or resource-limited settings. Machine learning offers a practical way to support screening by learning patterns from patient records and generating fast predictions from measured clinical features. In this project, the main question is whether these models can remain useful when they are moved from a conventional PC to a low-cost embedded device such as the Raspberry Pi.

II. Related Work

Previous studies have shown that common machine learning models can achieve strong CKD prediction accuracy. The reviewed work by Pal et al. reported high performance with classifiers such as logistic regression, decision trees, and support vector machines, with additional gains from bagging-based ensembles [6]. However, most prior emphasis is on classification accuracy rather than resource-aware deployment. The gap addressed in this report is the practical comparison between a general-purpose computer and an embedded platform, with attention to runtime and memory requirements in addition to predictive quality.

III. Methodology

A. Dataset and Preprocessing

The study used a publicly available CKD dataset containing 400 patient records. The features include common clinical measurements such as blood pressure, albumin, blood urea, serum creatinine, sugar-related indicators, and red blood cell information. Missing values were handled with mean or median imputation for numeric fields, categorical values were encoded, and numerical features were normalized before model training.

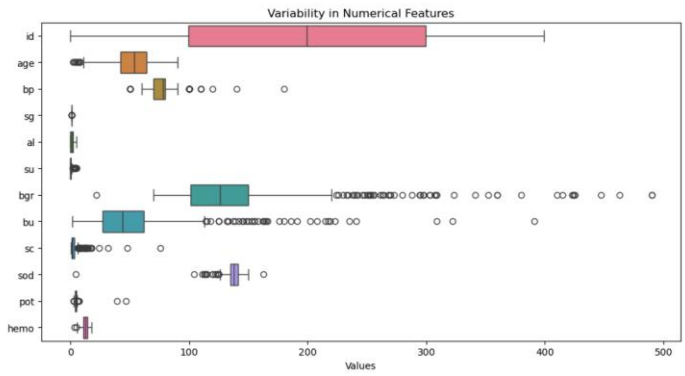
B. Model Training and Evaluation

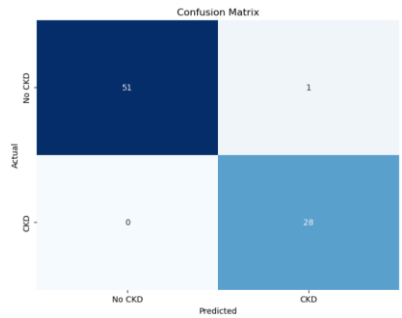
The processed dataset was divided into input features and target labels and then split into training and testing subsets. Logistic regression was used as the main baseline classifier, and the project also examined decision tree and neural network models. Performance was assessed with accuracy, confusion matrices, ROC-AUC, runtime, and memory usage. This combination of metrics makes it possible to compare both diagnostic quality and implementation cost on the PC and Raspberry Pi platforms.

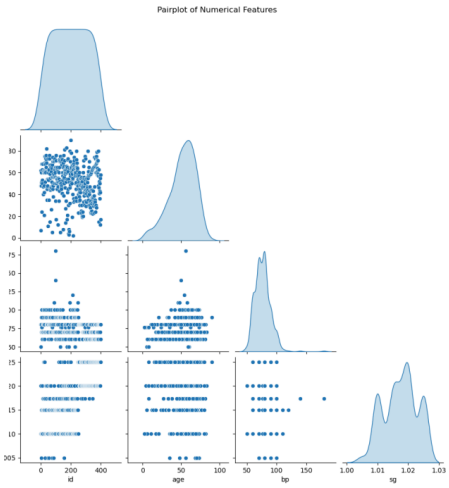
IV. Results and Discussion

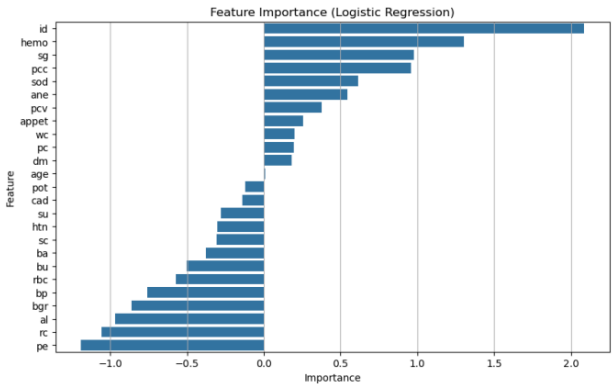
A. Logistic Regression Results

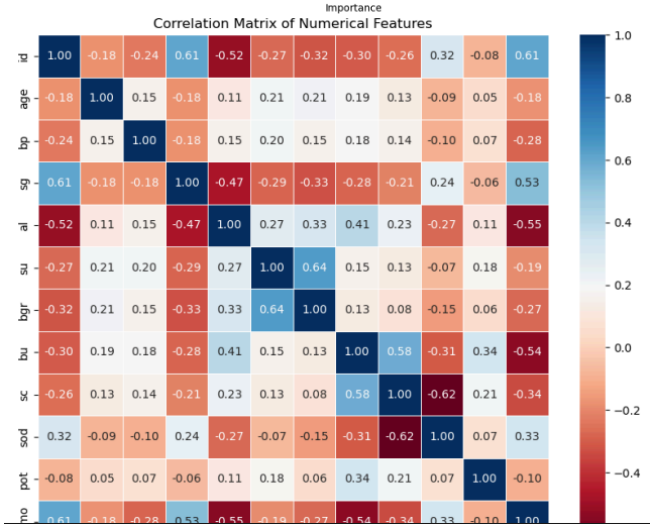
The logistic regression experiment achieved an overall accuracy of 87% with a ROC-AUC of 0.91. The confusion-matrix values reported in the original lab report were TP = 140, TN = 150, FP = 10, and FN = 20, which indicates balanced classification performance for CKD and non-CKD cases. Feature-level analysis identified serum creatinine and blood pressure as influential predictors.











B. *Neural Network Platform Comparison*

The neural network measurements reported in the original results were 87% accuracy, 4.0 runtime, and 80 memory units on the PC, compared with 89% accuracy, 1.5 runtime, and 10 memory units on the Raspberry Pi. These values show that platform changes affect runtime and resource use, but that embedded inference remains feasible for this task.

From an embedded-systems perspective, the key takeaway is that lightweight models remain attractive because they provide acceptable performance without large computational overhead. Logistic regression offered a strong balance between diagnostic accuracy and efficiency in the original report, while the neural network results suggest that higher-capacity models can still be considered when runtime and memory limits are acceptable. The work therefore supports the use of compact ML pipelines for affordable diagnostic support tools in underserved settings.

The study also has limitations. The dataset is relatively small, and the experiments were based on offline testing rather than real-world deployment with live sensor or clinical input. Future work should evaluate larger and more diverse datasets, test additional embedded platforms, and investigate further optimization methods for neural network inference.

Conclusion

This version reorganizes the original CME 465 lab report into an IEEE-style structure while preserving the main technical content. The reported experiments show that CKD prediction with machine learning is feasible on both a conventional PC and a Raspberry Pi. The overall results suggest that embedded deployment is practical when model complexity is matched to the available runtime and memory budget.

References

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